

same temperature but different masses will have different internal energies.

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A

$$\frac{1}{2}mc_{rms}^2 = \frac{3}{2}kT$$

$$\Rightarrow c_{rms}^2 \propto T$$

$$\frac{T}{2T} = \frac{c_{rms}^2}{c_{new rms}^2}$$

$$c_{new rms} = \sqrt{2}c_{rms}$$

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D